LAS 6292: Data Collection and Management

Emilio M. Bruna

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Preamble

This book is a manual for students in LAS 6292: Data Collection & Management. The course is designed for graduate students from any discipline – social sciences, humanities, biophysical sciences – and at all stages of their graduate program. It is an introduction to methods for collecting, organizing, managing, and visualizing both qualitative and quantitative data. Students will gain hands-on experience with best practices and tools.

Describe the different types of research data & the research data life-cycle -

Explain the need for and benefits of data management and sharing

Describe and implement best practices for data collection, storage, management, and sharing

Find, download, and analyze publicly available data from repositories

Carry out simple and reproducible data corrections and dataset organization

Describe public policies and agency requirements for data management and sharing

Articulate the major legal & ethical issues regarding the data collection, use, and storage

Create and Implement Data Management Plans in funder-specific formats

Identify and properly use tools for more efficient and secure data collection in the field

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Citation

This guide is a Quarto Book hosted on the Bruna Lab's Github site. Please suggest edits to the text, missing resources, or make suggestions for improvement either by pull request or by posting an issue on the course book's repository. For more information and tutorials for doing so, see **?@sec-manual**.

Part I.

Course Topics

1. Naming Conventions

1.1. Introduction

The first (and easiest) way to get started organizing your data is to use a simple, clear, and consistent system for naming your files (i.e., a "naming convention"). Using a well-thought out naming convention has a number of important benefits:

- Files will be easier to find
- You won't have to open files to see what they are
- Files are easier to sort
- Files are easier to share with collaborators (and for collaborators to use)
- It helps prevent accidentally overwriting or deleting files

1.2. How to name your files

Your file names should:

1. **Tell you about the contents of the file and allow you to uniquely identify it.** For instance (compare `'data 1998'` vs `'survey data'` vs `'survey data 1998'` vs `'small mammal survey data 1998'`). You can use things like the an acronym for the project, abbreviation for the study, location where collected, the name of the investigator, the year or month when collected, the type of data type, etc.
2. **Be as short as possible.** Many database systems (like MS One Drive) have limits on file name length. Use a max of 25 characters, and aim for less than that.
3. **Avoid using special characters and spaces** Don't use characters like `$ % ^ & # |` :. This is really important because software, automated data processing tools, web browsers, and computer operating systems use spaces and special characters for dividing up (parsing) strings of text, as well as other processes.
4. **Start with letters, never numbers.** Using numbers at the start of file names can lead to problems when they are read into stats software and other programs. Instead of `'2020_survey_responses'` use `'survey_responses_2020'`
5. **Help you quickly sort files chronologically or numerically**. Dates are very useful in file names because they can help identify the most recent files easily. You can *not* rely on the file date associated with the file on the computer! However, it is really important to use a clear format for dates, especially if collaborating internationally. I learned the hard way that not everyone reads `'census_data_3-4-2018'` the same way I do, so I started using `'census_data_4march2018'`. However, even this isn't ideal because of

how it sorts things in a folder. Now I now use this format: `'file_name_YYYYMMDD'` (`'census_data_20180403'`). You can also use `file_name_YYYY-MM-DD`. This is based on the international format for dates and times (ISO 8601). In addition to the benefit of sorting more easily, you'll never have to use the the dreaded `'_____.final.docx'` and `'_____.actual_final.docx'` in your file names ever again…you'll always know which is the last one. If the files don't require a date, but you want to maintain a sequence, use leading zeros to maintain sort order. `7` and `701` don't sort the way you expect them to, so use `007` and `701`.

6. **Use a consistent method of dealing with spaces and letter case.** . For a the spreadsheet with responses to a survey by 5 different people, don't use `'survey responses Florida'`. Instead use (i) `'survey_responses_florida'` or (ii) `'SurveyResponsesFlorida'`. Try to avoid mixing the two (e.g., `'survey_responses_Florida'`). By the way, (i) is known as `pothole` and (ii) is known as `camel`. Personally, I prefer pothole name is better because of it is easier to read and you never to remember what words get capitalized (because none of the do). Use underscore (`_`) or dashes (`-`) to separate meaningful parts of file names.

Once you have a system, write it down, print it out, and put it up somewhere you can easily refer to it. Remember -whenever possible make the names simple, informative, and unique. If you collected survey data and behavioral data in 2017, don't call them both `data_2017` even if they are in different folders. Call one `'behavior_obs_2017'` and the other `'survey_responses_2017'`).

Simple names also help you avoid problems with "system and software portability" - you might be working on a Mac but your collaborators are working on Linux or Windows, and the software made for each can read file names in different ways. File names that are simple, lower-case, and have no special characters are less software- and platform-dependent.

1.3. Tools & Resources

1.3.1. Bulk Renaming of Files

- Adobe Bridge
- Bulk Rename (Windows)
- Rename Master (Windows)
- File Renamer Basic (Windows, Free and Paid)
- Ant Renamer
- Renamer 5 (macOS)
- Renamer Lite
- PSRenamer
- Batch Renaming of Files in Windows: Post 1, Post 2
- Batch Renaming of Files in MacOS: Post 1, Post 2

1.3.2. Best Practices - Other Institutions

- Smithsonian Library

- Michigan Library
- Stanford Library
- Smithsonian

Part II.

Appendix 1: Useful Resources

2. Useful resources for Data Collection & Managment

2.1. R Programming

2.1.1. Essential

1. Hadley Wickham wrote a book on using the tidyverse and the online version is FREE. This is a phenomenal resource on using R to import, tidy, and visualize data.
2. Posit Cheat Sheets: help with commands for using the different `tidyverse` packages, RStudio shortcuts and tricks, help with R commands, and more. You definitely want the ones for Data Import, Work with Strings, Factors, Data Transformation, and Base R.
3. Where and How to ask for help
   - Hadley Wickham's advice on how to write a good reproducible example for getting help with R
   - how to post good questions on StackOverflow
   - The UF R-users listserv is *very* user friendly and a great place to post requests for help.

2.1.2. Tutorials and Books

1. R Essential Training: Wrangling and Visualizing Data
2. Software Carpentry: Using RStudio for Project Organization & Management
3. Swirl
4. R Bootcamp
5. Kieran Healy's *Data Visualization: a practical introduction* is my favorite introductory (yet super-comprehensive) book on data visualization with R. If you scroll down to the bottom of the page you can download the datasets and code used to make the figures in the book, which makes life much easier.
6. So. Many. Resources.
7. ROpenSci: tools for accessing, manipulating, and visualizing open data

8. How to clean messy data in R
9. The Ultimate Guide to Data Cleaning
10. Learning R
    Swirl

3. Specific Problems in Data Cleaning and Managemnt

1. Handling dates and times in R
2. Text Mining: `tidytext` package
3. Working with Qualtrics survey data with the `qualtRics` package
4. Optical Character Recognition (OCR): extract text from images: `tesseract` package
5. Extract text & metadata from pdf files: `pdftools` package
6. Image processing in R: the `magick` package

3.0.1. Advanced R Packages

1. `DataCurator` package: 'a simple desktop data editor to help describe, validate and share usable open data'.
2. RegExr: online tool to learn, build, & test Regular Expressions (RegEx / RegExp)
3. janitor (cleanup of file names, etc.)
4. `knitr` overview: reproducible documents with R
5. qualtRics
   ## Discipline-specific Resources
6. `historydata` package: Sample data sets for historians learning R. They include population, institutional, religious, military, and prosopographical data suitable for mapping, quantitative analysis, and network analysis.
7. *The Programming Historian* Website: wide range of topics, from text analysis to OpenRefine

3.1. Slide Presentations in R

1. Make slide presentations with R

3.2. Data Archives

Qualitative Data Repository

3.3. Text Extraction and Organization

Plan for extraction and organization

3.4. Form Design

Best Practice for Form Design

3.5. Data Security

UF Office of Information Security and Compliance
Cyber Safeguards for UF
UF IRB
UF Data Classification Policy
UF Office of Information Security and Compliance

3.6. Platforms for Organization & Collaboration

Open Science Framework